

SELECTED DISTRIBUTIONS OF RANDOM VARIABLES

Uniform Distribution of Discrete Random Variable

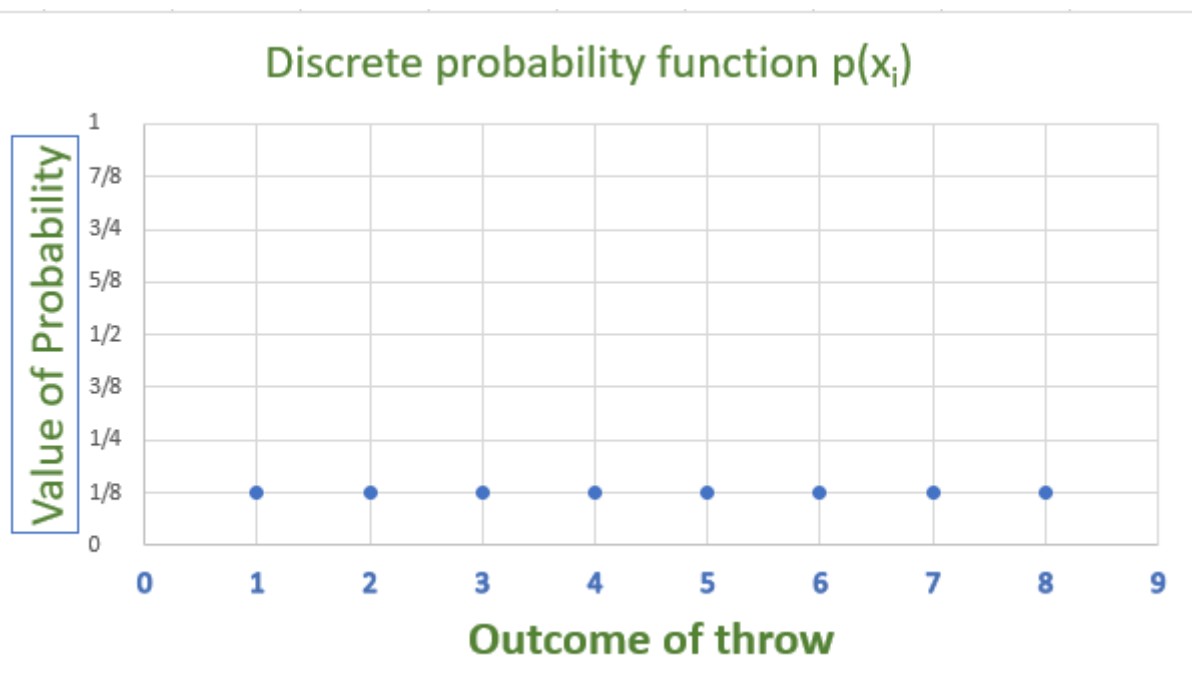
For all possible outcomes, i.e., values of a random variable, the probability is equally equal.

Example 1. Display probability function and distribution function for single throw of 8-sided dice.

Solution:

$$P(X=i) = 1/8 \text{ for } i = 1, 2, \dots, 8$$

The distribution of the probability function is uniform



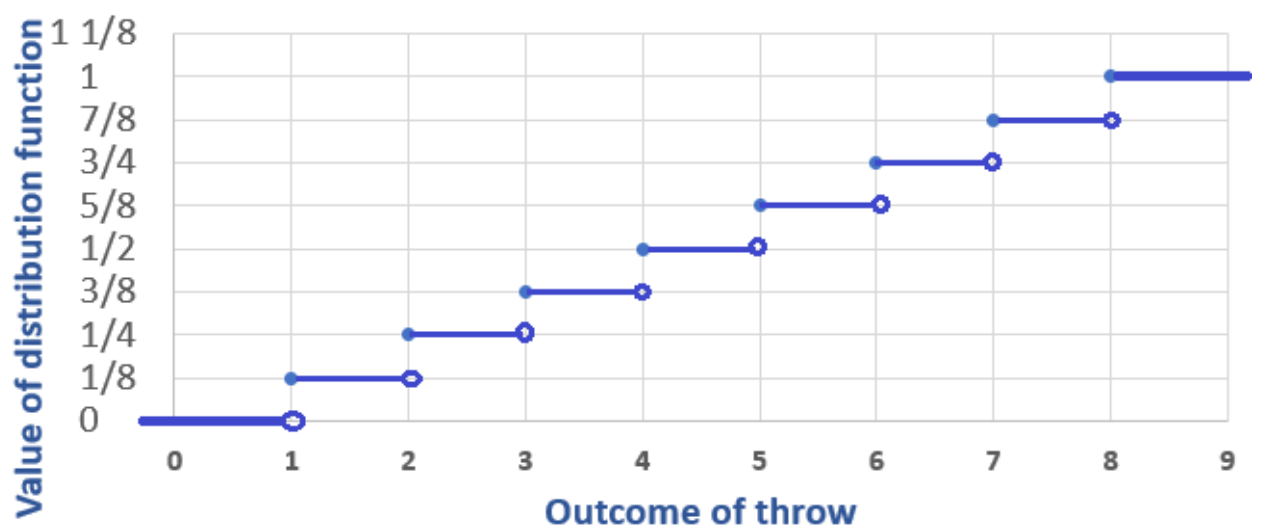
The discrete distribution function of a uniform distribution defined as

$$F(x_i) = p (X \leq x_i)$$

is a right-continuous linear step function

$$F(x_i) = \sum_{i=-\infty}^{x_i} p_i$$

Distribution function $F(x_i) = p (X \leq x_i)$



Example 2. Five percent of computer parts produced by a certain manufacturer are defective.

What is the probability that a sample of 20 parts contains more than three defective parts?

Solution:

We will use the opposite case, i.e., that fewer than four parts are defective in our sample of 20. Then the probability that the sample of 20 parts contains more than three defective parts is equal to

Geometric Distribution

Example 4. Quality control in a light bulb factory shows that the probability of producing a defective product is equal to $p = 0.1$.

What is the probability of finding a defective bulb in a batch of 100 bulbs produced in the consecutive n - draw, if a randomly selected *good bulb* is returned to the set of 100 bulbs?

Solution.

We use the geometric distribution. The probability of finding a defective bulb in the n^{th} inspection/draw

$$P(X=n) = p(1 - p)^{n-1}$$

where $n=1, 2, 3, \dots$

$$P(X=1) = p(1 - p)^{n-1} = 0.1 * 0.9^0 = 0,1$$

$$P(X=2) = p(1 - p)^{n-1} = 0.1 * 0.9^1 = 0.09$$

$$P(X=3) = p(1 - p)^{n-1} = 0.1*0.9^2 = 0.081$$

.....

$$P(X=100) = p(1 - p)^{n-1} = \dots$$

Zadej parametr	k= 100	(0≤k≤120)
Zadej parametr	p= 0.1	(0<p<1)
	Odeslat	

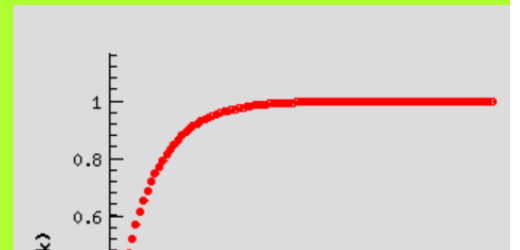
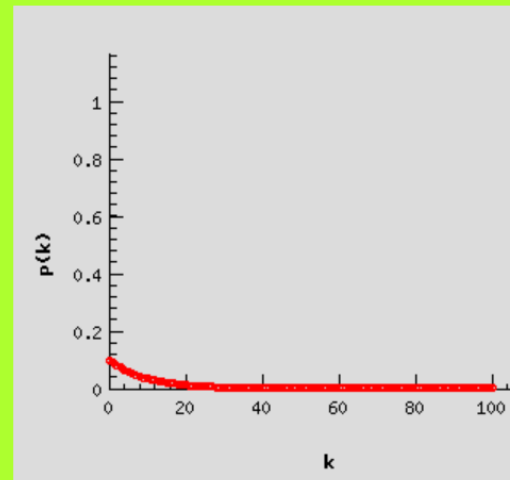
Vztahy

$P(X = k) = (1 - p)^k \cdot p$	$P(X \leq k) = 1 - (1 - p)^{k+1}$
$E(X) = \frac{1 - p}{p}$	$D(X) = \frac{1 - p}{p^2}$
$\gamma_1 = \frac{2 - p}{\sqrt{1 - p}}$	$\gamma_2 = 6 + \frac{p^2}{1 - p}$

Charakteristiky rozdělení

Střední hodnota	E(X)= 9
Rozptyl	D(X)= 90
Koeficient šikmosti	$\gamma_1 = 2.00277585144$
Koeficient špičatosti	$\gamma_2 = 6.01111111111111$

k	P(X=k)	P(X≤k)
0	0.1	0.1
1	0.09	0.19
2	0.081	0.271
3	0.0729	0.3439



Continuous Random variables.

Poisson distribution.

Example 4. The distribution of the number of days students are absent from compulsory classes during the semester is given by a probability function with parameter $\lambda = 2.5$.

Calculate the probability that a student will be absent less than 2 times during the semester.

Check the results using the calculator: <https://stattrek.com/online-calculator/poisson.aspx#Calculator>

Solution:

Absent less than 2 times means that

$$P(X < 2) = P(X=0) + P(X=1)$$

Applying expression of Poisson distribution we get

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda} \text{ for } k=0,1,2,\dots \lambda>0$$

$$P(X = 0) = \dots\dots\dots$$

$$P(X = 1) = \dots\dots$$

Exponential distribution.

Solution:

The compilation time distribution of a certain program is given by the density

$$f(x) = \begin{cases} a \cdot e^{-ax} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

with parameters $E(X)=1/a=5$ and $D^2(X)=1/a^2=25$

Calculate the probability that the compilation will take longer than 3 minutes.

Solution:

$$P(X>3) = 1 - P(X<3) = 1 - \int_0^3 \frac{1}{5} e^{-\frac{x}{5}} dx = 1 - e^{-\frac{x}{5}} \Big|_0^3 = e^{-\frac{3}{5}}$$

Using the calculator <https://www.omnicalculator.com/math/e-pow> 0,472er-x

we get

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Check the result using the calculator <https://www.integral-calculator.com/>

HW. Exponential Distribution cont.

Problem 6. Suppose you run a school cafeteria. The average number of customers is 15 students per hour.

What is the probability that you will have to wait no longer than 3 minutes for a customer to arrive?

Solution.

Hint. It will be practical to use 1 minute as the time interval. 15 students per hour means 15 students in 60 minutes, or 1 student every 4 minutes. Hence, parameter $a = \frac{1}{4} = 0.25$